

KOA MARIE GERVAIS NELLY

Lead AI Engineer & Data Architect | Founder @ Archi Cam AI

EXECUTIVE SUMMARY

Lead AI Engineer & Data Architect (Google Developer Program Member), leveraging Google Antigravity to design neuro-symbolic autonomous multi-agent architectures, Neo4j GraphRAG, and sovereign MLOps. Expert in AI ethics (transparency, SHAP explainability, EU AI Act compliance, and deterministic hallucination-free models). Founder of Archi Cam AI (official applicant for Google Africa Applied AI Lab), bridging civil engineering methodology, math rigor, and product vision.

FLAGSHIP AI PROJECTS

Archi Cam AI — *Agentic AI SaaS & 5D BIM*

- Official Applicant Google Africa Applied AI Lab. Built with Google Antigravity (Gemma 4 12B, Gemini, BAEL 91).
- Automated Excel BOQs in <45s (–99.2% time, MLflow $R^2=0.9872$) and 3D renders via Imagen 3 + ControlNet.

K1-MATHINFO (v3.0.0) — *Sovereign Multi-Agent AI, WikiSkills & OKF Certification*

- Sovereign DMI system (Univ. of Ngaoundéré): 470 theses, 19 M1 projects, 1,366 Neo4j nodes.
- 6 LangGraph agents orchestrated via WikiSkills (skills: critic anti-hallucination, Cypher optimizer), RRF $k=60$, SHA-256 No-LLM (100%).

Sovereign.BI Agentic — *Enterprise Security & Agentic BI*

- NL-to-SQL/Graph engine (PostgreSQL pgvector, Neo4j N10S, <5s latency) with ABAC guardrails + SHAP Sentinel.

Dataset Automator & VigieSahel — *MLOps Pipeline & Climate AI Impact*

- Dataset Automator: Google Antigravity (Google Cloud Hackathon) — TabFM, BigQuery DataFrames, EU AI Act.
- VigieSahel: –35% crop loss, +14d epidemic forecast (XGBoost $R^2>94\%$, Supabase, MLflow).

PROFESSIONAL EXPERIENCE

AI Lead & Data Science Consultant — *2025 – Present*

Independent Projects & Enterprises | Douala, CM

- Ethical sovereign AI, Neo4j GraphRAG pipelines, PostgreSQL, high-dimensional EDA, executive dashboards.

Aviation Security Officer (AVSEC) — *2018 – Present*

CCAA (Cameroon Civil Aviation Authority)

- Threat assessment, secure access control, regulatory compliance audits (ICAO Annex 17).

EDUCATION & CERTIFICATIONS

M.Sc. in Applied Artificial Intelligence — *2025 – 2027 [In Progress]*

University of Ngaoundéré

- ML & Bayesian Statistics, Data Engineering & Neo4j, Computer Vision & Robotics, Ethics & Cybersecurity, Production MLOps.

B.Sc. in Civil Engineering (Building Option) — *2015 – 2016*

ISTDI / IUC Douala

- Structural calculations (BAEL 91), quantity surveying, and construction project management.

HONORS & APPLIED AI RECOGNITIONS

- Google Africa Applied AI Lab (Accra, 2026):** Official candidacy — Archi Cam AI platform.
- Google Cloud #AllThingsAgentic Hackathon:** Dataset Automator v4.0 (Google Antigravity, TabFM, WIT, bigframes, MCT).



CONTACT & PROFILES

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🏆 Google Developer Program

AI & LLM STACK

Google Antigravity IDE ■■■■■
LangGraph & WikiSkills ■■■■■
Google Gemma 4 (12B) ■■■■■
Gemini 2.5 / 1.5 Pro ■■■■■
Google TabFM (Tabular) ■■■■■
Firebase Genkit ■■■■■
Neo4j GraphRAG — Agent K1

DATA & GRAPHS

Neo4j / Cypher Graph ■■■■■
PostgreSQL / pgvector ■■■■■
Google BigQuery ■■■■□
Supabase Realtime ■■■■□
Pandas / NumPy ETL ■■■■■

DEV & MLOPS

Python 3.11+ / MLOps ■■■■■
FastAPI / Next.js 14 ■■■■□
MLflow & Data Drift ■■■■□
SHAP Sentinel Audit ■■■■■
Docker & Streamlit ■■■■■
IfcOpenShell (5D BIM) ■■■■□

ETHICS, SECURITY & AUDIT

- AI Ethics & Anti-Hallucination
- Quorum 4-Eyes Governance
- OKF v0.2 SHA-256 No-LLM
- EU AI Act Compliance (RSASSA)
- AVSEC (ICAO Annex 17)

LANGUAGES

French — Native / Fluent
English — Technical / Pro

KEY ASSETS

- Dual Competence: AI & Civil Eng.
- Aviation Security & Crisis (CCAA)
- Mathematical Rigor & Guardrails